

Task 1 Write-Up: Minute-Trend Adversity Analysis

1. Market Behavior and Adversity Trends

By tracking the adversity percentage (trades resulting in an LP loss) from $\tau = 5$ to 30 seconds, we categorize client flow based on their micro-trends and the critical 50% loss threshold.

Safe / Reverting Flow (Clients A, B, C):

- **Client A (Mean-Reverter):** Adversity peaks at $\tau = 15$ (42.18%) and then *decreases* to 41.71% by $\tau = 30$. Their trades cause only a temporary market impact that quickly fades.
- **Client B (Plateau):** Adversity completely flatlines after $\tau = 15$, barely moving from 43.30% to 43.29% at $\tau = 30$. They possess zero long-term directional edge.
- **Client C (Slow Creep):** Displays a linear, slow climb (adding $\sim 0.5\%$ -1% per 5 seconds), ending at 45.92%. The velocity is too slow to ever cross the 50% danger zone.

Toxic / Alpha Flow (Clients D, E, F):

- **Client D (Late Bloomer):** Starts safely at 43.3% ($\tau = 5$) but steadily climbs, breaching the 50% loss threshold between $\tau = 20$ and $\tau = 25$ (hitting 50.61%).
- **Client E (Mid-Horizon Threat):** Experiences a sharp 3.5% jump early on, crossing the 50% threshold rapidly at exactly $\tau = 15$ (50.91%). Any position held longer than 15 seconds becomes a mathematical loss.
- **Client F (Hyper-Toxic):** Extremely dangerous flow. Adversity surges by 14.8% over 30 seconds, pushing us into net-loss territory almost instantly by crossing 51.66% before $\tau = 10$.

2. Strategic Action Plan

To maximize our LP performance, we must execute an asymmetric quoting strategy:

1. **Capture:** Offer aggressive (tight) spreads to Clients A and B to easily absorb their non-directional, profitable flow.
2. **Monitor:** Provide standard liquidity and spread widths to Client C.
3. **Defend:** We must drastically widen spreads and throttle volume for Clients D, E, and especially F. Because Client F crosses our loss threshold in under 10 seconds, our spread must be wide enough to absorb their immediate adverse price swings.